

Deepak Mallampati

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US-based (Kent, OH) - F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production AI services and customer-facing technical work. Delivered Manga Lens to the Chrome Web Store with multi-provider abstraction across four AI vision providers (Claude, GPT-4o, GPT-4.1, Ollama local), unified YOLOv8 real-time detection at ~30% lower inference latency, and FastAPI/Docker inference services that cut post-deployment defects ~30%. Comfortable reading unfamiliar Python and JavaScript/Node, debugging real-time pipelines, and translating field feedback into roadmap input - exactly the loop a Forward Deployed Engineer owns from first API call to production.

CORE COMPETENCIES

Forward Deployed Engineering • Customer Activation • Multi-Provider AI Abstraction • Real-Time Inference • Latency Optimization • Python • JavaScript / Node • FastAPI / Flask • Docker • REST APIs • LLM / RAG / Agentic • Structured Outputs & Evals • Observability & Structured Logging • Stakeholder Communication

WORK EXPERIENCE

Progress Solutions Inc. - Jr. AI/ML Engineer Trainee

Jul 2025 - Present

USA | Healthcare Technology, AI Consulting, SaaS Healthcare

- Shipped **FastAPI/Flask** RESTful inference services packaged with **Docker**, structured logging, and load simulation; **~30% fewer post-deployment defects** - the same reliability loop an FDE owns at customer handoff.
- Built **RAG systems** for clinical knowledge retrieval with recursive semantic chunking and transformer embeddings; **~35% retrieval precision gain**, **>30% irrelevant retrieval cut**, **>90% retrieval-grounded alignment** in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; **~25% agent response stability gain**.
- Delivered predictive ML pipelines (scikit-learn, XGBoost) for patient no-show / care engagement / support prioritization; **15-20% recall gain** on high-risk categories with precision held **>90%** via class weighting and threshold calibration.
- Wrote stakeholder-facing system-limitation docs and evaluation audit trails - the same architecture-discovery cadence FDEs run with customers; HIPAA-conscious data governance throughout.

Energy Solutions International (Emerson) - Database & DevOps Performance Engineer (Intern)

Jun 2022 - Apr 2023

Hyderabad, India | Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; **~20% query execution time cut**, **~25% retrieval latency cut**.
- Built **Jenkins CI/CD** for schema and stored-procedure deployments with structured release validation and rollback checkpoints; **>30% deployment-error reduction**, **~35-40% faster release cycles**.
- Designed Grafana + SQL Server DMV dashboards tracking long-running queries, deadlocks, CPU/IO contention; **~25% incident recurrence cut**, **~30% faster root-cause resolution**.

Suvidha Foundation - Machine Learning Engineer

Jan 2022 - Mar 2022

Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization + highlight system** across 5,000+ sessions with timestamp-aligned clip extraction; **60-70% manual review time cut**, **~85% alignment** with human-curated key moments.
- Shipped document Q&A with RAG (semantic chunking + embedding retrieval); **~30% hallucination reduction**, **>85% contextual accuracy**; deployed as a Flask API with a lightweight web interface.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation (*shipped, Chrome Web Store*)

Manifest V3 extension with content scripts, service workers, and `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) behind a unified abstraction. **Provider-specific payload handling** (WebP for cloud, JPEG for Ollama to avoid CUDA crash) - the exact "debug a customer's WebSocket/audio path under load" reflex an FDE leans on. 7-day translation cache, per-domain selector configs for 29 sites, privacy policy, narrowed host permissions.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; **~30% inference latency cut**. Sliding-window confidence aggregation to suppress blink-driven false positives (**~25% reduction**), adaptive frame skipping, and NMS tuning for stable real-time operation over OpenCV video capture.

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating Perplexity Sonar for interpretation and Replicate image models for 16:9 visuals. Inserted **intermediate structured prompt transformation layers** - **~30% contextual alignment gain**, **~25-30% first-pass image success gain** over naïve direct-prompt orchestration.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation, ANN similarity search for duplicate detection, explainable risk scoring with audit-ready reasoning traces.

Agentic Pixel Character Synthesis Engine (*ongoing*)

Stable Diffusion + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** decomposing prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, Node-style content scripts, C++ (Arduino), PHP

LLM & GenAI: Multi-provider LLM abstraction, RAG, agentic workflows, structured outputs, prompt engineering, evals, guardrails, grounding, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Real-Time / Multimodal: YOLOv8, OpenCV, viewport-slice screenshot pipelines, audio/video analytics, ControlNet, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, structured logging, load simulation, RESTful APIs, cloud-ready deployment

Customer-Facing: Stakeholder docs, system-limitation docs, evaluation audit trails, HIPAA-conscious governance